PAPER 2 1 hour

Essay

Answer **four** *questions* **only.**

All questions carry equal marks.

All working must be clearly shown. Marks will **not** *be awarded for correct answers without corresponding working.*

1. (*a*) A fair die and a fair coin are thrown together **once.**
   (i) Write down the set of all possible outcomes.
   (ii) Find the probability of obtaining a prime number and a tail.

   (*b*) The map of a field is drawn to a scale of 1 : 100. If the width and area of the field on the map are 8 cm and 88 cm² respectively, find in m^2, the area of the actual field.

   (*c*) Copy and complete the 3 × 3 magic square such that the sum of the numbers in **each** row, column and diagonal is equal to 21.

10	3	
	7	

2. (*a*) Given the vectors $\mathbf{p} = \begin{pmatrix} m+3 \\ 2-n \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 3m-1 \\ n-8 \end{pmatrix}$ and $\mathbf{p} = \mathbf{q}$, find the values of m and n.

   (*b*) A man shared an amount of money between his children Baaba and William in the ratio 6 : 5. Baaba received GH¢1,200.00.
   (i) Find the total amount shared.
   (ii) William invested his share in an account at the rate of 20 % simple interest per annum for 2 years. Find the total amount in his account at the end of the 2 years.

3. (*a*) Simplify: $3\sqrt{50} + 2\sqrt{45} - \sqrt{2} + \sqrt{5}$.

   (*b*) A wire of length 38 cm is bent into the shape of a rectangle whose length is 7 cm **more than** the width. Find the area of the rectangle.

   (*c*) If 15 % of the length of a rope is 720 metres, find half of the length of the rope.

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4. (*a*) Using a ruler and a pair of compasses **only**, construct ΔPQR such that angle $PQR = 90°$, $|PQ| = 5.5$ cm and $|QR| = 8$ cm.

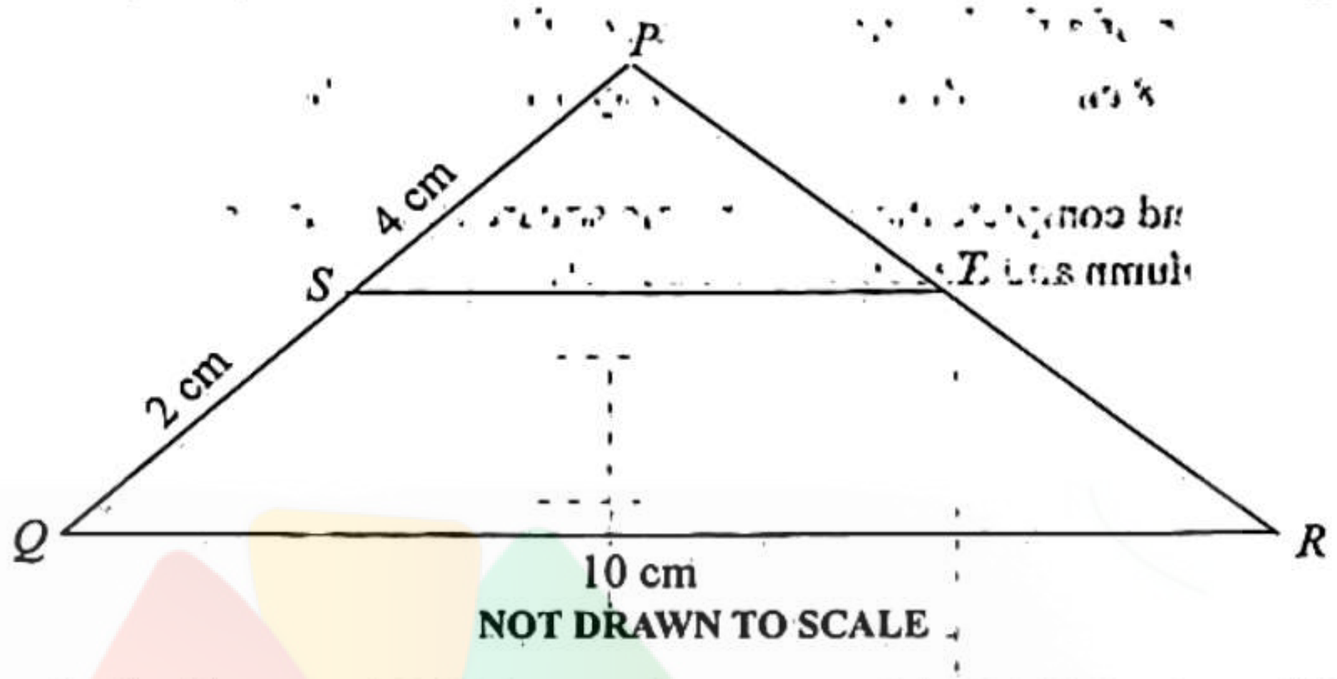
(*b*) Construct a perpendicular of PR from Q.

(*c*) Locate M, the intersection of the perpendicular and PR.

(*d*) Measure :
(i) $|MR|$;
(ii) $|QM|$.

(*e*) Calculate, correct to the **nearest** whole number, the area of triangle QMR.

5.



(*a*) In the diagram, ΔPQR is an enlargement of ΔPST. $|PS| = 4$ cm, $|QS| = 2$ cm and $|QR| = 10$ cm.
(i) Find the length of $\overline{ST}$.
(ii) If $|PQ| = |PR|$, find the area of ΔPQR.

(*b*) The total area of a school compound is $900\frac{1}{2}$ m^2. The school has Administration and Classroom block, Library, School Park, Roads and Walkways. The areas of the Administration and Classroom block, Library and School Park are $300\frac{1}{4}$ m^2, $200\frac{1}{2}$ m^2, and $120\frac{1}{8}$ m^2 respectively. Find the area covered by Roads and Walkways altogether.

6. (*a*) Copy and complete the table for the relation $F = \frac{9}{5}C + 32$.
where F and C are degrees Fahrenheit and degrees Celsius respectively.

°C	0	5	10	15	20	25	30
°F	32				68		

(*b*) Using a scale of 2 cm to 10 units on the vertical axis (°F) and 2 cm to 5 units on the horizontal axis (°C), draw a linear graph for the relation.

(*c*) Use the graph to find the temperature in degrees celsius when $F = 55$ degrees.

(*d*) Interpret the slope of the relation.

END OF ESSAY TEST

Turn over

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